RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 2 (Lambda)
Topic(s): Complex Numbers and Sequences and Series (Questions)

1 A sequence $u_1, u_2, u_3, \ldots$ is given by

$$u_1 = 2 \text{ and } u_{n+1} = \frac{3u_n - 1}{6} \text{ for } n \geq 1.$$

(i) Find the exact values of u_2 and u_3, and the value of u_{10} to 5 decimal places. [3]

(ii) It is given that $u_n \to l$ as $n \to \infty$. Showing your working, find the exact value of l. [2]

(iii) Describe the behaviour of the sequence. [1]

2 A sequence of numbers $u_1, u_2, u_3, \ldots$ has a sum S_n where $S_n = \sum_{r=1}^{n} u_r$.

(i) It is known that the sum of the first three terms of the sequence is 17 and $\sum_{n=1}^{3} S_n = 30$.

Furthermore, the third term of the sequence is twice the first term of the sequence. Find the first three terms of the sequence. [3]

(ii) A constant k is subtracted from each term of the sequence so that the first three terms of the modified sequence are now three consecutive terms of a geometric progression with a common ratio greater than 1.
Find the value of k and the common ratio of the geometric progression. [3]

3 A sequence $a_1, a_2, a_3, \ldots$ is such that $a_n = 2a_{n-1} - 3n + K$, where K is a constant, and $n \geq 2$.

(i) Given that $a_1 = 2$ and $a_2 = 4$, find K, a_3 and a_{18}. [2]

It is known that the nth term of this sequence is given by $a_n = p(2^n) + qn + r$, where p, q and r are constants.

(ii) Find p, q and r. [3]

(iii) Find $\sum_{n=1}^{N} a_n$ in terms of N. [2]

4 A sequence $u_1, u_2, u_3, \ldots$ is defined by the following recurrence relation,

$$u_1 = 1, \quad u_{n+1} = \frac{1}{2}u_n + 1 \text{ for all } n \in \mathbb{Z}^+.$$

Another sequence $v_1, v_2, v_3, \ldots$ is defined by $v_n = u_n - 2$ for all $n \in \mathbb{Z}^+$.

(i) Find a recurrence relation of the form $v_{n+1} = \mathrm{f}(v_n)$. Hence show that the sequence $v_1, v_2, v_3, \ldots$ is a geometric progression with common ratio $\frac{1}{2}$. [3]

(ii) Find the limit of the sequence $v_1, v_2, v_3, \ldots$ and that of the sequence $u_1, u_2, u_3, \ldots$ when $n \to \infty$. [3]

(ii) Find the sum to infinity of the series $v_1 + v_2 + v_3 \ldots$. [1]

(iv) Find the sum of the first n terms of the sequence $u_1, u_2, u_3, \ldots$. Determine whether the series $u_1 + u_2 + u_3 + \ldots$ converges. [3]

5 Bank *A* offers a study loan to students enrolled in an undergraduate course of study. The key features of the loan are:

- interest-free during the course of study,
- fixed monthly interest of 0.3% upon graduation,
- minimum monthly repayment of \$100.

Ali decides to take a study loan of \$50000 from Bank *A* on 1 January 2021 at the start of his 3-year undergraduate course.

(a) During his course of study, Ali pays \$200 at the end of January 2021 and on the last day of each subsequent month, he pays \$10 more than in the previous month. Thus on 28 February 2021, he pays \$210 and on 31 March 2021, he pays \$220, and so on. How much does Ali owe the bank at the end of his course of study? [2]

(b) Bank *A* charges interest on any outstanding amount of the loan on the first day of each month, starting on the month right after a student graduates.

(i) Upon graduation, Ali immediately found a job and decides to pay \$900 to bank *A* on the last day of each month, starting on the month right after he graduated. Show that Ali owes the bank \$34121 (to the nearest dollar) on the last day of the 3rd month after he graduated. [2]

(ii) Use the formula for the sum of a geometric progression to find an expression for the amount owed by Ali on the last day of the *n*th month after he graduated. Hence find in which month Ali pays off his study loan. [5]

(iii) Find the total amount of interest that Ali paid. [2]

(iv) If Ali decides to pay off his study loan within 3 years upon graduation, i.e. at the end of December 2026, what is the minimum amount, to the nearest dollar, that he needs to pay per month after graduation? [3]

6 A geometric progression has first term a and common ratio r, and an arithmetic progression has first term a and common difference d, where a and d are non-zero. The sums of the first 2 and 4 terms of the arithmetic progression are equal to the respective sums of the first 2 and 4 terms of the geometric progression.

(i) By showing $r^3 + r^2 - 5r + 3 = 0$, or otherwise, find the value of the common ratio. [5]

(ii) Given that $a < 0$ and the *n*th term of the geometric progression is positive, find the smallest possible value of n such that the *n*th term of the geometric progression is more than 1000 times the *n*th term of the arithmetic progression. [3]

7 When a ball is dropped from a height of H m above the ground, it will rebound to a height of eH m where $0 < e < 1$. The height of each successive bounce will be e times of that of its previous height. It is also known that the time taken between successive bounce is given by $t = 0.90305\sqrt{h}$ where h is the maximum height of the ball from the ground between these bounces. We can assume that there is negligible air resistance.

A ball is now dropped from a height of 10 m from the ground. Let t_n be the time between the nth and $(n+1)^{\text{th}}$ bounce.

(i) Show that the total distance travelled by the ball just before the nth bounce is $\dfrac{10(1+e-2e^n)}{1-e}$. [3]

(ii) Show that t_n is a geometric sequence. State the common ratio for this sequence. [3]

(iii) Find in terms of e the total distance the ball will travel and the time taken when it comes to rest. You may assume that between any two bounces, the time taken for the ball to reach its maximum height is the same as the time it takes to return to the ground. [3]

8 **Do not use a calculator in answering this question.**

(i) Solve the equation $z^2 + 1 + \left(2\sqrt{2}\right)\mathrm{i} = 0$, leaving the roots in the form $x + \mathrm{i}y$, where x and y are real. [4]

(ii) One root of the equation $3s^3 - 8s^2 + 31s - 18 = 0$ is $1 + \left(2\sqrt{2}\right)\mathrm{i}$. Find the other roots. [3]

Deduce the exact roots of the equation $3p^6 + 8p^4 + 31p^2 + 18 = 0$. [4]

9 Two complex numbers w and u satisfy the equations

$$w^* - 2\mathrm{i}u = 8 \quad \text{and} \quad (2\mathrm{i} - 1)w + 2u^* = 4.$$

Find w and u, giving your answers in the form $x + \mathrm{i}y$, where x and y are real. [4]
Hence find $|u^* - w|$ and $\arg(u^* - w)$. [2]

10 Given $u = 1 + \sqrt{3}\mathrm{i}$ and $v = 1 - \mathrm{i}$ are roots of the equation $z^4 + az^3 + bz^2 + cz + d = 0$, where a, b, c and $d \in \mathbb{R}$,

(a) find the values of a, b, c and d. [4]

The complex numbers $\mathrm{i}u$ and v^* are represented by the points P and Q respectively on an Argand diagram.

(b) Find the area of triangle OPQ. [3]

11 Do not use a calculator in answering this question.

The complex number w is such that $w = \mathrm{i} - \sqrt{3}$.

(i) Find $|w|$ and $\arg(w)$ in exact form. [2]

(ii) Represent w, $-w$ and $2-w$ in the same Argand diagram, illustrating clearly the geometrical relationship between them. [3]

(iii) Hence, find $\arg(2-w)$ exactly. [2]

12 Do not use a calculator in answering this question.

It is given that $z = \sqrt{3} - \mathrm{i}$ and $w = 2\mathrm{i}z$.

(a) Find the modulus and argument of z. [2]

O, A, B and C represent the complex numbers 0, z, $z+w$ and w respectively.

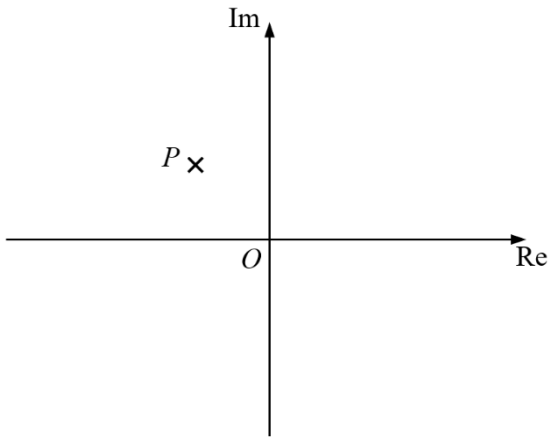
(b) Illustrate these four points on a single diagram. [2]

(c) Deduce the modulus and argument of w. [2]

(d) Identify the shape of the quadrilateral $OABC$ and state its area. [2]

13 Do not use a calculator in answering this question.

The complex number z has modulus 1 and argument θ, where $\frac{\pi}{2} < \theta < \pi$, and the complex number w is given by $w = \mathrm{i}\sqrt{3}z$. The point P on the Argand diagram represents z.



(a) On the copy of the Argand diagram with origin O in the Printed Answer Booklet, plot the points Q and R to represent w and $z-w$ respectively. Show clearly the geometrical relationship between the points P, Q and R. [3]

(b) Find the area of the quadrilateral $ORPQ$. [1]

It is given that $\theta = \frac{3\pi}{4}$.

(c) Find z in the form $x + y\mathrm{i}$, where x and y are real numbers. [2]

(d) Show that $z - w = k\left[\left(\sqrt{3}-1\right)+\left(\sqrt{3}+1\right)\mathrm{i}\right]$, where k is a constant to be determined. [2]

(e) Hence show that $\tan\frac{5\pi}{12} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$. [1]